

CNN (Convolutional Neural Network)

A Convolutional Neural Network (CNN) is a deep learning algorithm mainly used for image classification and computer vision tasks. It automatically detects important features like edges, textures, and shapes from images.

A CNN consists of multiple layers:

1. Convolutional Layer

Applies filters (kernels) to extract features
Produces feature maps
Detects edges, patterns, textures

2. Activation Function (ReLU)

Applies non-linearity

3. Pooling Layer

Reduces image size (downsampling)
Types:
Max Pooling
Average Pooling

4. Flatten Layer

Converts 2D feature maps → 1D vector

5. Fully Connected Layer (Dense)

Performs classification

6. Output Layer

Uses Softmax for multi-class classification

Working of CNN

- Input image is passed to convolution layer
- Features are extracted using filters
- Activation function applied
- Pooling reduces dimensions
- Flatten converts data
- Dense layer predicts output class

Applications of CNN:

- Image classification
- Face recognition
- Medical image analysis
- Self-driving cars
- Video surveillance
- Handwriting recognition
- Object detection